

Assignment

ETL – Data Extraction

Question 1: Describe different types of data sources used in ETL with suitable examples.

Answer:

Data sources are the places from where data is collected for processing in an ETL (Extract, Transform, Load) system.

The common types of data sources are:

1. Relational Databases

These databases store data in tables with rows and columns.

Examples: MySQL, Oracle, SQL Server, PostgreSQL.

Example: Customer information stored in a MySQL database.

2. Flat Files

Flat files store data in a simple file format.

Examples: CSV, TXT, and TSV files.

Example: A CSV file containing student names, roll numbers, and marks.

3. Excel Files

Excel files are commonly used by organizations to store and manage data.

Example: An Excel file containing monthly sales records.

4. APIs

APIs allow applications to request data from another application or service.

Example: A weather API providing current temperature and weather information.

5. Web and Cloud Sources

Data can also be extracted from websites and cloud-based services.

Examples: Google Sheets, cloud storage, and web services.

Thus, ETL can extract data from many different sources and combine it into a common system for analysis.

Question 2: What is data extraction? Explain its role in the ETL pipeline.

Answer:

Data extraction is the process of collecting data from different sources such as databases, files, APIs, and applications.

ETL stands for:

- **E – Extract**
- **T – Transform**
- **L – Load**

Extraction is the **first step of the ETL pipeline**.

Role of Data Extraction:

1. It collects data from different sources.
2. It identifies the required data.
3. It retrieves the data from the source system.
4. It transfers the extracted data to the transformation stage.
5. It helps combine data from multiple sources.

Example:

An organization extracts customer data from a database and sales data from CSV files. The data is then transformed and loaded into a data warehouse.

Therefore, data extraction provides the raw data required for the remaining ETL process.

Question 3: Explain the difference between CSV and Excel in terms of extraction and ETL usage.

Answer:

CSV and Excel are both commonly used as data sources in ETL, but they have different features.

CSV	Excel
CSV stands for Comma-Separated Values.	Excel is a spreadsheet file format.
Stores data mainly as plain text.	Can store data, formulas, charts, and formatting.
Usually contains one table.	A workbook can contain multiple sheets.
Simple and lightweight.	More feature-rich and complex.
Easy to process with ETL tools.	Requires Excel-compatible processing.
Suitable for large simple datasets.	Better for structured business spreadsheets.

Example:

A company may use a CSV file to exchange daily sales data, while an Excel workbook may contain sales data divided into separate sheets for different regions.

Question 4: Explain the steps involved in extracting data from a relational database.

Answer:

The following steps are generally used to extract data from a relational database:

Step 1: Identify the Data Source

First, identify the database and the required tables.

Example: MySQL database containing a **Customer** table.

Step 2: Establish a Connection

The ETL tool or program connects to the database using the required credentials.

Step 3: Identify Required Data

Determine which tables, columns, and records are required.

Step 4: Write a Query

An SQL query is used to retrieve the required data.

Example:

```
SELECT Customer_ID, Name, City  
FROM Customer;
```

Step 5: Extract the Data

The database executes the query and returns the required records.

Step 6: Validate the Extracted Data

Check whether the data is complete, correct, and in the expected format.

Step 7: Send Data to the Next ETL Stage

The extracted data is passed to the **Transformation** stage for cleaning and processing.

Thus, database extraction allows ETL systems to efficiently retrieve required data from structured databases.

Question 5: Explain three common challenges faced during data extraction.

Answer:

Three common challenges during data extraction are:

1. Large Data Volume

Extracting a very large amount of data can take a lot of time and system resources.

Example: Extracting millions of customer records from a database.

Solution: Use batch processing, incremental extraction, and optimized queries.

2. Data Quality Problems

Source data may contain missing values, duplicate records, or incorrect information.

Example: A customer record may have a missing phone number.

Solution: Validate and clean the data during the ETL process.

3. Different Data Formats

Data can come from different sources and formats such as CSV, Excel, databases, and APIs.

Example: One system stores dates as `DD-MM-YYYY`, while another uses `YYYY-MM-DD`.

Solution: Standardize the data format during transformation.

Therefore, proper planning and validation are important for successful data extraction.

Question 6: What are APIs? Explain how APIs help in real-time data extraction.

Answer:

API stands for **Application Programming Interface**. An API allows one software application to communicate with another application and request data or services.

APIs are useful for **real-time or near-real-time data extraction** because an ETL system can request the latest available information directly from an application or service.

How APIs help:

1. They provide data directly from applications or services.
2. They allow automated data extraction.
3. They can provide updated information whenever a request is made.
4. They reduce the need for manual data collection.
5. They allow different systems to communicate with each other.

Example:

A weather application can use a weather API to request the latest temperature and weather conditions.

Question 7: Why are databases preferred for enterprise-level data extraction?

Answer:

Databases are preferred for enterprise-level data extraction because they are designed to store and manage large amounts of structured data efficiently.

Main reasons:

1. **Large Data Handling:** Databases can store and process millions or billions of records.
 2. **High Performance:** SQL queries can retrieve required data efficiently.
 3. **Security:** Databases provide authentication and access-control mechanisms.
 4. **Data Consistency:** Database rules help maintain accurate and consistent data.
 5. **Concurrent Access:** Multiple users and applications can access data at the same time.
 6. **Reliability:** Databases provide backup and recovery features.
 7. **Scalability:** Database systems can grow as the organization's data increases.
-

Question 8: What steps should an ETL developer take when extracting data from large CSV files (1GB+)?

Answer:

When extracting data from very large CSV files, such as files larger than 1GB, an ETL developer should use efficient methods to avoid memory and performance problems.

Important steps:

1. Do Not Load the Entire File into Memory

Read the CSV file in smaller **chunks or batches**.

2. Check the File Structure

Verify the delimiter, column names, data types, encoding, and number of columns.

3. Process Data in Batches

Divide the file into manageable portions and process each portion separately.

4. Select Only Required Columns

Extract only the columns needed for the ETL process.

5. Validate the Data

Check for missing values, duplicate records, invalid formats, and corrupted rows.

6. Handle Errors

Log problematic records separately instead of stopping the complete extraction process.

7. Use Parallel Processing When Appropriate

For very large files, processing can sometimes be divided into multiple tasks to improve performance.

8. Monitor Performance

Monitor memory usage, processing time, and the number of records processed.

9. Store Intermediate Results

If required, save processed batches temporarily so that the process can resume after a failure.

Example:

Instead of loading a 2GB CSV file completely into memory, the ETL developer can read it in batches, validate each batch, transform it, and load it into the target database.